Comments to the Author

The manuscript proposes sensor-aware masked-reconstruction pretraining for gas sensor arrays, with a shared dilated 1-D CNN, learned per-channel identity embeddings, and cross-sensor attention, followed by low-label adaptation for concentration regression. The leakage-control protocol is careful and above the norm for this literature, and the authors report inconvenient results rather than suppressing them. Major revision is needed for the following reasons.

Major

1. The premise is not tested. Pretraining reuses controlled exposures with labels withheld, so the labels already exist. The real cost in this field is generating the exposure, not annotating it. Either pretrain on genuinely unlabeled data (ambient, disjoint concentration regime, or a different array) or drop all language implying reduced calibration cost.
2. The "Best policy" column in Table I selects between frozen and fine-tuned transfer on the test partition (frozen for CO, fine-tuned elsewhere), contradicting the statement in Section V.B. All four abstract gains inherit this. Pre-register one policy, or select on the adaptation-only validation split, and recompute the abstract.
3. Three seeds is not enough. NH3 and H2 scratch and transfer distributions overlap at about one standard deviation, no hypothesis tests are given, and Table III has no error bars and is non-monotonic in several places (CO 20 vs 24 pairs, ethylene 5 vs 10 pairs). Use at least 10 seeds with error bars on Tables III and IV.
4. The multi-fold results in Section V.D are the most credible in the paper and are currently subordinated. They show smaller gains (16.4 percent for NH3, 14.5 percent for ethylene) and roughly double the absolute error, with an across-fold standard deviation of 17.81 ppm. The paired bootstrap intervals all exclude zero and are a defensible result. Promote this analysis to primary and demote the single-split tables.
5. Drift partition 10 (146.95 plus or minus 8.71 against 142.78 plus or minus 1.13) is a null, not a 2.8 percent gain, and should be stated as such. This section also needs a constant-predictor control emitting the per-gas mean from the adaptation set.
6. The sensor-aware claim needs a sensor-agnostic masked self-supervised baseline (for example TST, TS2Vec, or PatchTST). Internal ablations do not establish it. The feature representation given to Extra Trees and XGBoost is never stated, and a PLSR or SVR chemometric baseline is missing.
7. Equation (16) selects calibration pairs using the concentration labels it is meant to economize, so it is a design-of-experiments choice, not label selection. A random-selection control is also absent.
8. Learned per-channel identity embeddings bind the encoder to one array and preclude cross-device transfer. State this limitation, and ideally test pretrain on array A, adapt on array B.
9. The private array is not characterized: no sensing material, operating temperature, sampling rate, baseline convention, or humidity and temperature conditions. Humidity in particular cannot be omitted for MOX sensors. Also, only about 68 of roughly 430 samples per exposure are used, with no justification for the window placement, and the 0 to 500 ppm NH3 range does not match the environmental monitoring framing.

Moderate
CO performance is weak (R squared 0.776, Q0.9 of 118.82 ppm on a 0 to 533 ppm scale) and should be acknowledged as such. The conformal coverage failure is structurally guaranteed given the non-exchangeable design and needs either a shift-aware variant or an explicit statement of the violated assumption. Table IV needs error bars, and the 25-fold spread in the sensor identity effect between H2 and NH3 needs explanation or attribution to noise. MacroMAE in ppm is dominated by high-concentration analytes; report per-gas normalized error. The log-target Huber optimum is biased for linear-space MAE and the per-gas scaling is estimated from only 16 records. Please report parameter counts against effective sample size, clarify whether repeated exposures appear in the test partition, add scratch Q0.9 values, and either reimplement one competing method or evaluate on a published split. Seventeen references is thin.

Presentation
Section V.B states "increase MAE significantly" where "decrease" is meant, which is a sign error in the sentence introducing the main result. Please correct "overal", "trainning", "comparision", and "Multif-fold". The Keywords field contains only "SYST". Tables I and II largely duplicate each other. Fig. 3 labels are unreadable at print size. Fig. 2 is clear and should be kept.